Hermann Hankel — Zur Geschichte der Mathematik in Alterthum und Mittelalter (1874)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Hermann Hankel, who died at thirty-four with the history of mathematics itself unfinished on his desk. He had planned it in two stages — one volume for a wider circle, then a larger account for specialists — and death stopped even the first stage. What survived was a sheaf of worked-out sections among his papers, and his father, Wilhelm Gottlieb Hankel, published those sections unchanged in 1874. This presentation is dedicated to the son who wrote it, and to the father who brought it into print.

The author's own terms

Hankel called his guiding rule the *Prinzip der Permanenz der formalen Gesetze*, the principle of the permanence of formal laws: when a number system is extended, the formal laws of the old system are to keep ruling in the new.

The story

Hermann Hankel was born on 14 February 1839 at Halle, son of the physicist Wilhelm Gottlieb Hankel. The family moved to Leipzig in 1849, when his father took the chair of physics there, and at the Nicolai Gymnasium the

schoolboy read the ancient mathematicians in the original Greek. He entered the University of Leipzig in 1857, went to Göttingen in 1860 as Riemann's student, and worked with Weierstrass and Kronecker in Berlin in 1861. Appointed extraordinary professor at Leipzig in spring 1867, he was ordinary professor at Erlangen by that autumn, and in 1869 he accepted the chair at Tübingen for his last four years. His *Theorie der complexen Zahlensysteme* (1867) presented the permanence principle, and MacTutor, following the *Dictionary of Scientific Biography*, records that he was the first to recognise the significance of Grassmann's long-neglected writings. The Hankel transform, matrix, and functions still carry his name. He died at Schramberg on 29 August 1873, aged thirty-four.

The work and what survives

Zur Geschichte der Mathematik in Alterthum und Mittelalter (Leipzig: B. G. Teubner, 1874; (iv) + 410 pages) carries two appendices, a fragment on Euclid at page 381 and a sketch of the mathematics of the Chinese at page 405. Its editor's preface, signed by his father at Leipzig on 3 October 1874, states that the worked-out sections are published unchanged, covering antiquity and the Middle Ages except classical-period geometry, the Euclid fragment joined as an appendix. After the opening Division and the pre-scientific chapters, it runs to Renaissance algebra, breaking off at page 380. The *Dictionary of Scientific Biography* records a Hildesheim reprint in 1965. MacTutor's verdict stands: the historical writings "contain many errors, yet they are filled with brilliant insight."

What this project holds is plain. The project folder holds one text file only, of 8,742 words: fresh English of the editor's preface, the Division, and the Numbers and Number-Words section, printed pages 1 to 22 only, translated from the Internet Archive OCR text, not line-collated against page images, and not yet past the expert review gate, a historian of mathematics and a reader of nineteenth-century German.

Its English fortunes

The corrected finding stands at this strength: **no complete or book-length English translation of Hankel's history was found after research; isolated**

passages have been Englished inside other scholars' works. Two such passages are pinned. A passage translation beginning at page 155, on Plato and instrumental constructions, is hosted as a PDF at the College of the Holy Cross; it names no translator, date, or publication status, so it stands as an unattributed rendering, not a published edition. Pages 164 to 165 are cited as Englished in Ulrich Libbrecht's *Chinese Mathematics in the Thirteenth Century* (Dover, 2005), at page 218. That second pin rests on the citing reference alone; Libbrecht's page was not page-checked, and the finding carries that bound.

What completing it would take

Completing the 1874 volume would mean fresh English for the remaining printed pages 23 to 380, from numerals through Renaissance algebra, together with both appendices. The part already translated needs page-image collation, resolving Greek words and a Breton numeral surviving only in degraded OCR, and the whole must pass the review gate named above. The bound is firm. The classical Greek geometry was never among the worked-out sections, beyond the Euclid fragment, and the larger specialist history was never written. Those cannot be completed, only marked; any completion can claim only the whole of what his father published in 1874.

Sources

- Project folder: ~/workspace/rare-and-obscure-books/books/hankel-geschichte-der-mathematik/hankel-geschichte-der-mathematik.txt (preface and contents)
- https://mathshistory.st-andrews.ac.uk/Biographies/Hankel/ (MacTutor)
- https://siteproxy.duoduoqaq.workers.dev:443/https/mathshistory.st-andrews.ac.uk/DSB/Hankel.pdf (Crowe, DSB)
- https://old.maa.org/press/periodicals/convergence/mathematical-treasure-hankels-history-of-mathematics (MAA Convergence)
- https://www.Gutenberg.Org/files/31061/31061-pdf.pdf (Cajori)
- http://mathcs.holycross.edu/~little/Derek/Hankel.pdf (passage from p. 155)
- https://en.wikipedia.org/wiki/Diophantus_and_Diophantine_Equations (citing reference for Libbrecht, p. 218; not page-checked)